

求极限

$$\lim_{x \rightarrow 0} \frac{\tan \tan x - \sin \sin x}{\tan x - \sin x}$$

首先由泰勒展开式得到

$$\tan x - \sin x \sim \frac{1}{2}x^3$$

$$\begin{aligned}\tan \tan x &\sim \tan x + \frac{1}{3}\tan^3 x + \frac{1}{3}\left(\tan x + \frac{1}{3}\tan^3 x\right)^3 \\ &\sim \tan x + \frac{2}{3}\tan^3 x\end{aligned}$$

$$\begin{aligned}\sin \sin x &\sim \sin x - \frac{1}{6}\sin^3 x - \frac{1}{6}\left(\sin x - \frac{1}{6}\sin^3 x\right)^3 \\ &\sim \sin x - \frac{1}{3}\sin^3 x\end{aligned}$$

那么

$$\begin{aligned}\tan \tan x - \sin \sin x &\sim \tan x + \frac{2}{3}\tan^3 x - \sin x + \frac{1}{3}\sin^3 x \\ &\sim \frac{1}{2}x^3 + x^3 = \frac{3}{2}x^3\end{aligned}$$

因此

$$\lim_{x \rightarrow 0} \frac{\tan \tan x - \sin \sin x}{\tan x - \sin x} = 3$$